Voiceprint

Interpretable AI Voice Screening

For a fraud category with no defense — built on acoustic forensics, not a black box.

Problem Voice clones are undetectable by ear	**Solution** Acoustic signal analysis + plain-English verdicts
Differentiator Auditable reasoning — not a mystery score	**Model** B2B API/SDK — embedded forensic layer

⚠ THE PROBLEM

Voice-Clone Scams Are a Fast-Growing, Unsolved Fraud Vector

62%

Deepfake Attack Rate

Of organizations reported a deepfake-related attack in the past year

3s

Time to Clone

Seconds of audio now sufficient to convincingly impersonate any voice

High-Value Targets

CEO Fraud

Impersonated executives authorizing wire transfers

Elder Scams

Family voice impersonation targeting seniors

Call Centers

Bypassing voice authentication at scale

⊗ No accessible, real-time way for ordinary people to verify if a voice is real. **The defense is missing.**

THE SOLUTION

Interpretable Forensic Screening — Not a Black Box



Upload or Record

Submit a voice clip directly from browser or mobile — no specialized hardware, no install. Seconds to start.



Acoustic Signal Processing

Not just a neural net. Praat/librosa-style measurement of pitch, harmonic ratios, silence floor, and breath patterns.



Plain-English Evidence Cards

"Digital-zero silence detected, no natural breath turbulence." The reasoning is explicit — not a confidence score.



Auditable Verdict

Every conclusion traces back to measurable acoustic features. Judges, regulators, and users can trust the chain of reasoning.

> **Credibility through transparency.** Judges and users can follow the logic — they don't have to take the model's word for it.

TARGET SEGMENTS

Anyone Who Answers a Phone

Individuals

Family and elder-fraud protection. No technical knowledge required — browser-accessible verdict in seconds.

Businesses

Wire-transfer and CEO-fraud verification before high-value actions are authorized.

Call Centers & Fraud Teams

Real-time screening integration via API — embedded in existing fraud workflows without workflow disruption.

Journalists & Researchers

Audio authenticity verification for investigative work and academic research on synthetic media.

No specialized hardware needed — works from a browser or mobile SDK. Zero friction to first verdict.

TECHNICAL IMPLEMENTATION

Grounded in Acoustic Forensics — Not Assumption

Feature Extraction Layer

Pitch Jitter & Shimmer

Micro-variation in frequency and amplitude — suppressed in synthetic voices

Harmonic-to-Noise Ratio

Measures organic vocal noise structure absent in TTS output

Silence Floor Cleanliness

Digital-zero silence is a TTS artifact — natural speech has noise floor

Breath & Pause Naturalness

Human breathing patterns within pauses; absent or artifically uniform in clones

Classifier & Validation

Logistic Regression

Interpretable thresholds — weights are readable, not hidden in a deep net

Calibrated Outputs

Scores map to probability estimates, not opaque confidence values

Tested Against Real TTS Engines

ElevenLabs, LuvVoice, and others — iterative threshold refinement on real adversarial examples

Praat/librosa Measurement

Industry-standard acoustic analysis tools; reproducible and peer-validated methodology

BUSINESS MODEL

A Forensic Layer for a Fraud Category with No Consumer Defense Today

B2B API / SDK Licensing

Not a standalone consumer app. Voiceprint is positioned as an embedded forensic layer — dropped into existing products via API call or mobile SDK.

Why B2B First

- Lower go-to-market friction than direct consumer distribution
- Clearer, contractual revenue path through enterprise licensing
- Embeds in existing fraud workflows — no behavior change required from end users

Integration Points

Telecom

Call-screening at the carrier layer before connection

Banking & Fintech

Voice-based fraud verification before high-value transaction authorization

Enterprise Security

CEO and executive impersonation screening in internal comms

Consumer Safety Apps

SDK embedded in third-party safety or caller-ID applications

ROADMAP

From Acoustic Signals to a Broader Authenticity Layer

1

Phase 1 — Now

Clip-based browser screening. Feature extraction pipeline. Evidence card UI. Tested on ElevenLabs, LuvVoice.

2

Phase 2 — Training Expansion

Broader TTS engine coverage. Real-world recording conditions: compression artifacts, codec degradation, background noise.

3

Phase 3 — Real-Time Streaming

Live-call analysis replacing post-hoc clip submission. In-call verdict for call center and telecom integration.

4

Phase 4 — Multimodal

Video deepfake cues. Synchronized audio-visual authenticity checks. On-device mobile SDK for consumer apps.

Scalable foundation — each phase extends the same interpretable acoustic core. The explainability architecture scales to multimodal without redesign.

HONEST LIMITATIONS

What We'd Harden Next — No Hype, Just Facts

Dataset Expansion

Current thresholds calibrated on a small test set. Need broader validation across a wider range of TTS engines, speaker demographics, and real-world recording conditions before production deployment.

Pacing-Dependent Feature Confound

Digital-silence detector is confounded by speech pacing on short clips. **Fix identified:** measure silence only within detected pause windows, not across the full clip. Ready to implement and validate.

Live-Call Streaming

Current architecture is post-hoc (clip-based). Real-time in-call mode requires streaming pipeline — architecture scoped, not yet built.

Explainability Testing

Evidence cards have not yet been validated with real end-users. Must confirm plain-English explanations remain accurate and comprehensible as the model evolves.

Why We're Telling You This

Forensic and security tools earn credibility by being explicit about failure modes — not by hiding them. Every limitation above has a known path forward.

Known

Every limitation is already diagnosed — not discovered post-launch

Actionable

Each has a specific fix scoped and ready to execute

Honest

We know what we know and what we don't — that's the forensic standard

> *Credibility is built on honesty about what we know and what we don't.*